GazeTextVLM - Gaze + Text Guided Token Pruning for Efficient VLM Decoding

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Abstract

Vision-language models (VLMs) are increasingly used in embodied and egocentric settings, where users ask questions about their current visual environment in real time. However, autoregressive decoding remains expensive because visual tokens must be retained in the KV-cache throughout generation. We study whether inference-time visual token pruning can improve efficiency while preserving reasoning quality, using human gaze and textual relevance as pruning signals. We evaluate several pruning strategies with Qwen2-VL-2B-Instruct on the EgoGazeVQA benchmark, measuring multiple-choice accuracy, captioning quality, and chain-of-thought reasoning under pruning. Our results suggest that pruning up to 50% of visual tokens preserves reasoning performance relative to the unpruned baseline, while improving throughput on memory-bandwidth-limited hardware. These findings suggest that substantial visual redundancy exists in current VLM decoding pipelines and that aggressive token pruning can reduce inference cost without significantly degrading reasoning quality. Code is available on `https://github.com/juliagontijo/gazetextpruneqwen/tree/main`

1 Introduction

Vision-language models have become a standard interface for visual reasoning, supporting scene description, visual question answering, and grounded instruction following. Their utility is especially compelling in wearable and embodied settings, where a user queries a model about their immediate environment in real time. In augmented reality systems this interaction is naturally egocentric: the model receives first-person video input, and the user expects a fast response about ongoing actions or objects in the scene.

This expectation creates a trade-off between capability and latency. Visual encoders convert frames into hundreds of tokens that are retained in the KV-cache throughout autoregressive decoding, and for multi-frame inputs this cost compounds quickly. Reducing the KV-cache footprint by pruning irrelevant tokens at an early decoder layer is a natural target for efficiency gains, though whether those gains materialise in practice, and whether any pruning signal is better than random, remain empirical questions.

Egocentric settings provide a signal unavailable in standard VLM pipelines: gaze. When a user wearing an eye-tracked device asks a question, their fixation point reflects spatial attention, meaning it captures what they are looking at and not necessarily what they are reasoning about. Gaze is a plausible cue for spatial localisation tasks ("where is the card?"), but may be uninformative or even distracting for questions that require reasoning about cause, sequence, or object relationships beyond the fixation region. We therefore analyze whether gaze improves, degrades, or does not effect on pruning quality. Alongside gaze, the text prompt itself encodes relevance: a question such as "What is happening on the TV?" naturally identifies which objects should matter most. Text-guided pruning retains tokens with high attention weight from question-conditioned rater tokens and is a natural complement to gaze-guided pruning. A hypothesis this research explores is that combining both signals produces a better pruning score than either alone, improving reasoning performance over individual strategies.

We evaluate four inference-time visual token pruning strategies: random, gaze-guided, text-guided, and a linear combination, using Qwen2-VL-2B-Instruct on EgoGazeVQA, a benchmark of egocentric video clips with causal, spatial, and temporal multiple-choice questions. The consistent pattern across conditions is that pruning up to 50% of visual tokens does not significantly degrade accuracy or reasoning quality relative to an already weak baseline, suggesting that visual redundancy in VLM decoding pipelines is substantial. In exploring the tradeoff between pruning aggressiveness and reasoning capability, we aim to characterise the potential of token pruning as a tool for improving inference efficiency in egocentric VLM deployments.

2 Related Work

Efficient VLM inference and token pruning. Recent work has shown that the cost of VLM inference is driven in large part by the number of visual tokens passed to the language model. SparseVLM is a particularly relevant reference because it proposes text-guided sparsification of visual tokens, using the prompt to estimate which parts of the visual input are most relevant for generation [Zhang et al., 2025]. This line of work is important for our setting because it frames pruning as an inference-time selection problem rather than a full model redesign, and it suggests that language-conditioned relevance scores can preserve quality while reducing computation.

Gaze as a relevance signal for multimodal models. A separate line of work studies how human gaze can improve alignment between visual inputs and model predictions. VOILA-A (*Aligning Vision-Language Models with Userś Gaze Attention*) is directly relevant because it treats gaze as a meaningful supervision or guidance signal for VLM behavior [Yan et al., 2024]. More broadly, gaze-aware multimodal systems argue that user attention can reveal which objects, interactions, or regions are salient from the human perspective, especially in embodied or assistive applications. This makes gaze a compelling candidate for test-time relevance estimation even when the end goal is efficiency rather than alignment alone.

Gaze-grounded VLMs in egocentric settings. Prior gaze-language models are also relevant because they demonstrate that gaze contains information about user intent, scene understanding, and task focus. In particular, GazeVLM-style approaches use gaze within a trained modeling pipeline, showing that eye movements can improve grounding and prediction in multimodal tasks [Mathew et al., 2025]. Our motivation differs from this training-centric perspective: rather than using gaze only as a learned supervision signal, we are interested in whether gaze can help decide which visual information should be retained at inference time.

Egocentric question answering benchmarks. Finally, datasets such as EgoGazeVQA make this problem empirically meaningful by combining egocentric video, gaze traces, and question-answering tasks [Peng et al., 2025]. These benchmarks reflect exactly the kind of use case where efficient decoding matters: a first-person stream contains substantial redundancy, the userś attention is observable, and the question often concerns only a subset of the scene. Taken together, the literature suggests a clear opportunity at the intersection of efficient VLM inference, gaze-aware modeling, and egocentric QA.

3 Method

In order to study the inference-time visual token pruning as a mechanism for reducing autoregressive decoding cost in VLMs, we augment a frozen VLM with an inference-time pruning module inserted between the visual encoder and the language model decoder, as shown in Figure 1. The module scores each visual token using gaze and text signals and discards low-scoring tokens during decoder's prefil phase, reducing the KV-cache footprint for the remainder of the generation. We then evaluate how each scoring strategy - gaze-guided, text-guided, their linear combination, and random selection as a baseline - affects reasoning quality and decoding efficiency on egocentric VQA.

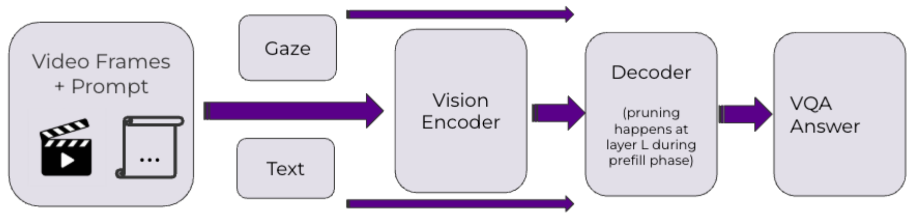

Figure 1: Overview of the GazeTextVLM pruning pipeline. Gaze and text signals are used to score visual tokens after encoding; pruning is applied once at decoder layer L, reducing the KV-cache for all subsequent layers.

Text-based relevance. For each visual token, we compute a relevance signal from the prompt-conditioned interaction between text and vision during prefill, following the text-guided token rating intuition introduced by SparseVLM [Zhang et al., 2025]. Intuitively, tokens that receive stronger task-related attention from the textual side are more likely to be useful for downstream decoding. This branch is intended to capture query relevance: if the prompt refers to a particular object, action, or region, the corresponding visual tokens should receive higher scores.

Gaze-based relevance. In parallel, we compute a gaze score over the visual token grid. The gaze input is given as a two-dimensional point in image coordinates, denoted by (g_x, g_y). We map this signal onto the visual token layout and assign higher scores to tokens closer to the attended location using a two-dimensional Gaussian defined on the visual grid and centered at (g_x, g_y). The Gaussian bandwidth is set to $\sigma = 0.15 \cdot \max(H, W)$, where H and W denote the height and width of the visual grid. This branch is intended to capture user attention: tokens near the gaze location are treated as more likely to contain information that the user considers important.

Score fusion and pruning. We combine the two signals into a single importance score for each visual token. Let s_{text} denote the text-based relevance score and s_{gaze} denote the gaze-based relevance score. Before fusion, both scores are min-max normalized to the range $[0, 1]$ so that the two branches are comparable in scale. We then define the fused score as

$$S = \alpha\, s_{text} + (1 - \alpha)\, s_{gaze},$$

where $\alpha \in [0, 1]$ controls the relative weight of language relevance and gaze relevance. Given these fused scores, we retain only the top-ranked visual tokens according to a user-specified keep ratio r, and prune the remainder.

Pruning. We retain the top-r fraction of tokens by fused score and discard the rest. Pruning is applied once at decoder layer L: all visual tokens are processed normally up to layer L, and only retained tokens propagate to subsequent layers, reducing the KV-cache footprint for the remainder of generation. Figure 2 shows the retained tokens for each strategy on the same frame and we can see how gaze-guided pruning concentrates retained tokens around the fixation region (TV screen) while text-guided pruning distributes them more broadly across task-relevant objects and combined pruning interpolates between the two patterns.

GRPO fine-tuning. Finally as a preliminary investigation, we also run a short GRPO fine-tuning experiment using MCQ accuracy as the reward signal, and evaluate the fine-tuned model under the same pruning conditions. The goal is not to claim that fine-tuning improves the model, but to assess whether reward-based training changes how robust the model is to visual token pruning.

3.1 Evaluation

We evaluate pruning across three complementary dimensions: MCQ accuracy, captioning similarity, and chain-of-thought reasoning quality assessed by an LLM judge.

MCQ accuracy. We report exact-match accuracy overall and by question type. Comparisons against the no-pruning baseline use McNemar's test on paired binary outcomes.

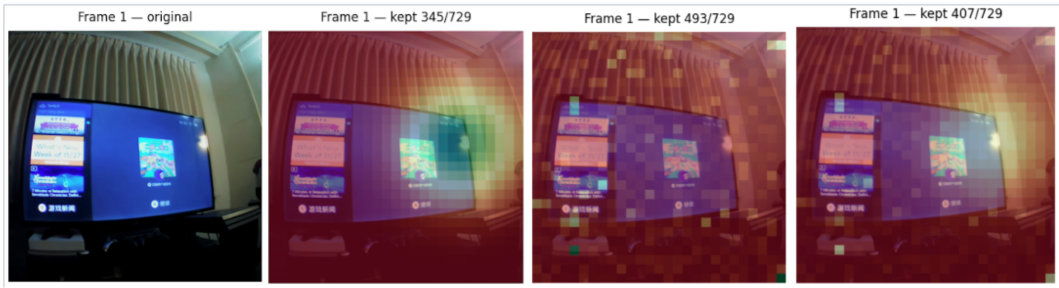

Figure 2: Visual token retention on a single EgoGazeVQA frame ($r = 0.5$, layer $L = 10$). From left to right: original frame (729 tokens), gaze-guided pruning (345/729 kept), text-guided pruning (493/729 kept), and combined pruning (407/729 kept). Retained tokens are shown in their original color and pruned tokens appear in red.

Captioning similarity. To test whether pruning degrades generative output beyond MCQ accuracy, we additionally prompt the model to produce a free-form scene description and measure similarity to the ground-truth Ego4D narration using RoBERTa-based sentence similarity.

LLM-as-judge. Exact-match accuracy cannot detect whether pruning degrades the reasoning chain that produced an answer. We therefore use an LLM-as-judge protocol [Zheng et al., 2023]: Qwen2.5-7B-Instruct rates each generated chain-of-thought on a 0-5 scale for overall reasoning quality and three sub-dimensions (scene grounding, question relevance, logical chain). We additionally run a pairwise preference evaluation in which the judge compares the pruned and unpruned CoT for the same sample and selects the better response.

4 Experiments

4.1 On-Device Latency: SmolVLM Prototype

Our starting point was a lightweight prototype using **SmolVLM** Marafioti et al. [2025] on two EgoExo4D Grauman et al. [2024] sequences running on CPU/MPS hardware. This pilot established that visual-token pruning delivers measurable on-device speedups before we scaled to a stronger reasoning model.

Pruning scheme. SmolVLM tiles each frame into a fixed grid (17 tiles per frame including one global tile). We score tiles by a 2D Gaussian centred on the per-frame gaze fixation point, retain the top-r fraction, and always keep the global tile for coarse scene context. Pruning happens after visual encoding but before the language-model decoder, so no architecture changes are required.

Table 1: On-device inference speedup (SmolVLM, CPU/MPS). Speedup is relative to the full-token baseline (17/17 tiles, 1088 tokens). BERTScore F1 measures output similarity to the unpruned model.

Keep ratio	Tiles kept	Tokens	Speedup	BERTScore F1
1.00 (baseline)	17/17	1088	1.00×	1.000
0.75	13/17	832	1.42–1.57×	0.959–0.967
0.50	9/17	576	1.54–1.64×	0.941–0.952
0.25	5/17	320	1.55–1.69×	0.938–0.947

Reducing from 1088 to 320 tokens yields a **1.55–1.69**× wall-clock speedup while BERTScore F1 remains above 0.94, confirming that gaze-guided pruning removes mostly irrelevant non-attended tokens. Speedup is not linear with the token reduction ratio because model-invocation and decoding-initialisation overheads are fixed costs but nonetheless the gains are substantial on memory-bandwidth-limited hardware.

These results motivate the remainder of this paper: latency is a real bottleneck on device, and gaze provides a cheap, always-available signal for reducing it.

Pruning Layer	Accuracy in Layer Sweep
5	50% (5/10)
10	70% (7/10)
15	60% (6/10)
20	60% (6/10)

Table 2: Layer ablation for visual token pruning.

The key open question is whether smarter pruning strategies preserve reasoning quality as the model is asked harder egocentric questions. Therefore we switch to **Qwen2-VL-2B-Instruct**, a stronger reasoning model, and evaluate on EgoGazeVQA with structured MCQ questions that demand causal, spatial, and temporal understanding.

4.2 Qwen2-VL Setup

Dataset. We evaluate on **EgoGazeVQA** Peng et al. [2025], egocentric video clips paired with per-frame gaze coordinates and five-way multiple-choice questions (MCQ) spanning three reasoning types: causal (why did the person do X), spatial (where is object Y), and temporal (what happened before/after Z). We sample 30 clips per question type (90 total) and extract 8 frames per clip at uniform temporal intervals.

Model and pruning. We use **Qwen2-VL-2B-Instruct** Wang et al. [2024] in `bfloat16` on a single NVIDIA A100. At a designated decoder layer ℓ, a fraction $(1-r)$ of visual tokens are dropped from the KV-cache, reducing the effective sequence length for all subsequent layers.

Hyperparameter selection. We selected layer 10 as the primary pruning layer because it achieved the highest accuracy in the layer ablation, as seen in Table 2. This suggests that layer 10 provides a good trade-off: it is early enough for pruning to reduce downstream computation, while still late enough that the model has formed sufficiently informative multimodal representations for reliable token ranking.

4.3 Multiple Choice Question Results

Table 3 reports accuracy and decode throughput across all pruning configurations ($n = 87$–90, A100). The central finding is that pruning up to 50% of visual tokens does not significantly degrade MCQ accuracy relative to the unpruned baseline (McNemar's test, all $p > 0.05$). Figure 3 visualises the overall and per-type accuracy breakdown.

Table 3: Full production results ($\ell = 10$, A100, $n = 87$–90 per condition). Accuracy columns show fraction correct. Judge is mean LLM-judge score (0–5). **Bold** = best per column among pruning methods. Baseline row shaded.

Method	r	Acc.	Causal	Spatial	Temporal	ms/tok	Judge
No pruning	—	0.47	0.71	0.27	0.43	38.8	1.98
Random	0.5	**0.50**	**0.77**	0.23	**0.50**	38.5	**1.97**
Text	0.50	**0.48**	0.67	**0.28**	0.48	37.6	1.89
Text	0.75	0.45	0.70	0.23	0.43	37.8	1.71
Gaze	0.50	0.45	0.61	0.27	**0.50**	37.2	1.71
Gaze	0.75	0.44	0.60	0.27	0.47	**37.0**	1.76
Combined	0.50	0.42	0.55	0.23	0.47	37.4	1.80

Several patterns emerge from the results. Spatial questions are uniformly hard across all conditions (0.23–0.28), close to the 0.20 chance level, suggesting this reflects an inherent limitation of the 2B model on fine-grained egocentric localisation rather than a pruning effect. Random pruning matches or exceeds structured methods overall (0.50 accuracy, 0.77 causal), which suggests that at $r = 0.5$ the model is robust to which tokens are removed — dropping half the visual context indiscriminately does not hurt and may reduce over-attention to irrelevant background regions. Gaze-guided pruning is the weakest on causal questions (0.61 vs. 0.71 baseline), consistent with the intuition that fixation location encodes spatial attention rather than causal relevance. Combined pruning at $\alpha = 0.5$ gives the lowest overall accuracy (0.42), suggesting that equal weighting dilutes both signals rather than combining their strengths.

On latency, no pruning method improves decode throughput on A100 at batch size 1, as the GPU is compute-bound rather than memory-bandwidth-bound at this scale. Meaningful speedups require memory-bandwidth-limited hardware, as demonstrated by our SmolVLM prototype in Section 4.1.

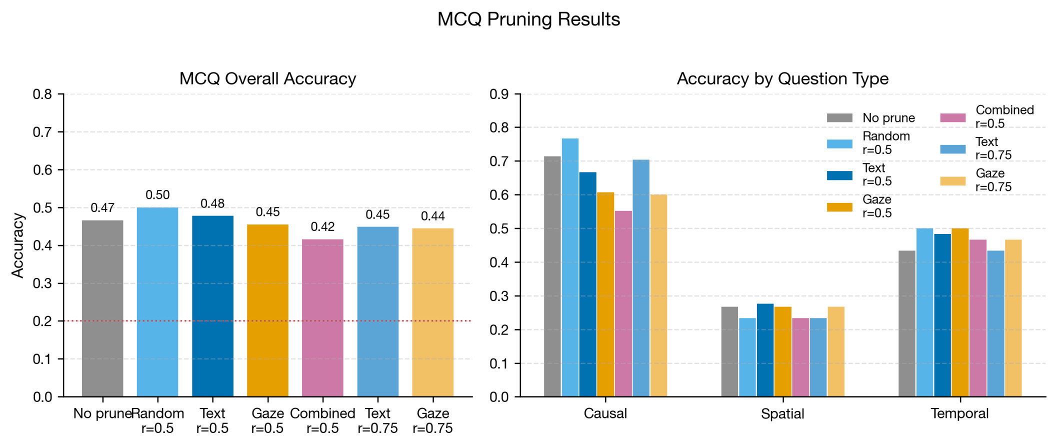

Figure 3: MCQ pruning results. Left: overall accuracy across pruning methods. Right: accuracy broken down by causal, spatial, and temporal question types. The dotted red line marks 5-way chance performance.

4.4 Reasoning Quality

Table 4: LLM-judge coverage sub-scores (0–5) and pairwise preference vs. no-pruning baseline ($n = 90$ for all).$\Delta = B - A$ wins. Positive means the pruned CoT was preferred more often.

		Coverage scores (0–5)					**Preference**			
Method	r	**Judge**	**Scene**	**Rel.**	**Logic**	**Final**	**A**	**B**	**Tie**	Δ
No pruning	—	**1.98**	**3.58**	2.88	**2.62**	**2.72**	—	—	—	—
Random	0.50	1.97	3.56	**3.08**	**2.66**	2.71	36	27	27	−9
Text	0.50	1.89	3.56	2.93	2.44	2.62	27	34	29	+7
Text	0.75	1.71	3.47	2.82	2.41	2.53	19	34	37	+15
Gaze	0.50	1.71	3.53	2.90	2.46	2.61	26	**40**	24	+14
Gaze	0.75	1.76	3.47	2.76	2.41	2.51	28	37	25	+9
Combined	0.50	1.80	3.51	2.86	2.48	2.56	33	33	24	0

Judge scores cluster tightly between 1.71 and 1.98 regardless of pruning method and the low absolute values reflect the difficulty of zero-shot egocentric VQA for a 2B model rather than any pruning effect. Scene grounding (mean=3.5) is the strongest sub-dimension across all conditions, while logical chain ($\approx$2.4–2.7) is the most sensitive to pruning: text at $r = 0.75$ shows the largest drop (2.41 vs. 2.62 baseline, $p = 0.056$, approaching significance). No pruning method significantly degrades reasoning quality (Wilcoxon signed-rank, all $p > 0.05$).

Figure 4 we can visualize the LLMjudge evaluations. We can see the preference results show a directional tendency for the judge to prefer pruned CoTs over the baseline for text-guided and gaze-guided pruning ($\Delta = +7$ and $+14$ respectively), though these differences do not reach statistical significance. Random pruning is the only condition where the baseline is preferred more often

($\Delta = -9$). Gaze-guided pruning earns the most outright B wins (40/90 at $r = 0.5$), possibly because spatially concentrated context shifts the CoT toward more object-grounded descriptions that the judge rewards. In

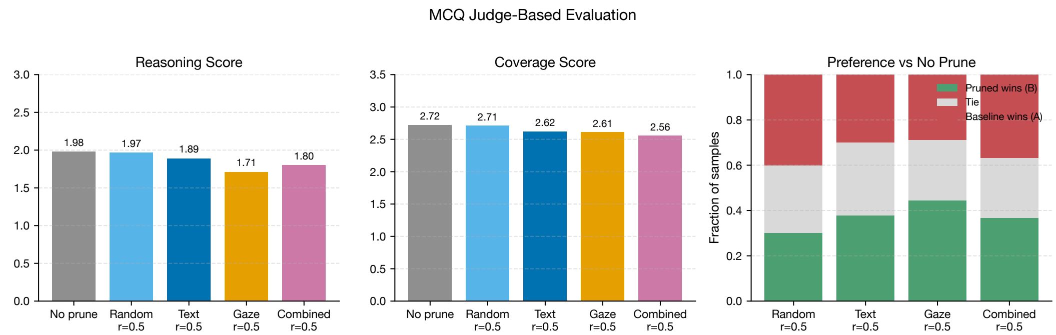

Figure 4: Judge-based evaluation of MCQ pruning results. Left: mean reasoning score. Middle: mean coverage score. Right: pairwise preference against the no-pruning baseline, where B indicates the pruned output was preferred.

4.5 Captioning

To test whether pruning degrades generative output beyond MCQ accuracy, we ran a captioning evaluation in which the model produces a free-form scene description for each clip with no multiple-choice prompt. We compute a similarity score between the generated caption and the ground-truth Ego4D narration (higher = closer to ground truth). We evaluate four conditions at $\ell = 10$, $r = 0.5$ ($n = 90$ each).

Table 5: Captioning results: similarity to ground-truth narration and decode throughput ($\ell = 10$, $r = 0.5$, A100, $n = 90$). **Bold** = best among pruning methods. Baseline row shaded.

Method	n	Similarity	ms/tok
No pruning	90	0.815	41.1
Random	90	0.815	39.2
Gaze	90	0.813	37.5
Text	90	**0.818**	**37.6**

The captioning results reinforce the MCQ findings. All pruned methods remain within 0.002 similarity points of the unpruned baseline, confirming that dropping half the visual tokens does not meaningfully degrade generative output quality. Text-guided pruning achieves a marginal improvement over baseline (0.818 vs. 0.815), consistent with its MCQ result, and yields a $\sim$8% throughput gain (37.6 vs. 41.1 ms/tok). Gaze-guided pruning shows the largest drop (-0.002), mirroring its weaker MCQ performance on causal questions: retaining only tokens near the fixation point may omit context relevant to scene-level descriptions.

4.6 GRPO Fine-Tuning: Proof of Concept

The pruning experiments above operate on a frozen base model. A natural follow-up question is whether fine-tuning changes the redundancy structure of visual tokens that is, whether a model trained to reason better about egocentric scenes becomes more or less robust to pruning. As a preliminary investigation, we fine-tuned Qwen2-VL-2B-Instruct using GRPO, treating MCQ accuracy as the reward signal, and evaluated the resulting model under the same pruning conditions as above. The run comprised 2 epochs over 13 gradient steps with a mean reward of 0.30 (max 0.75), the limited scale of this experiment serves as a proof of concept rather than a fully trained regime.

The results show early signs that GRPO reward shaping improves performance on reasoning-intensive question types: causal accuracy improved from 0.71 to 0.80 and temporal from 0.43 to 0.57 relative to the base model, suggesting the reward signal is meaningful even at this scale. However, spatial

Table 6: MCQ accuracy before and after GRPO fine-tuning ($r = 0.5$, $\ell = 10$). Base model rows are from Table 3; fine-tuned rows are evaluated on a 30-sample subset on different hardware and are not directly comparable on latency. **Bold** = best per column among pruned fine-tuned conditions.

Method	Acc.	Causal	Spatial	Temporal
Base model (no fine-tuning)				
No pruning	0.47	0.71	0.27	0.43
Gaze ($r = 0.5$)	0.45	0.61	0.27	0.50
Text ($r = 0.5$)	0.48	0.67	0.28	0.48
Combined ($r = 0.5$)	0.42	0.55	0.23	0.47
GRPO fine-tuned				
No pruning (FT)	0.43	0.80	0.08	0.57
Gaze (FT, $r = 0.5$)	**0.47**	**0.80**	**0.15**	**0.57**
Text (FT, $r = 0.5$)	0.43	0.80	0.08	0.57
Combined (FT, $r = 0.5$)	0.43	0.80	0.08	0.57

accuracy collapses to 0.08 for three of the four conditions, well below the 0.20 random chance level, suggesting the short training run over-fitted to causal and temporal patterns at the expense of spatial grounding which is a risk with RL fine-tuning on MCQ tasks at small batch and step counts. Pruning behavior on the fine-tuned model mirrors the base model: gaze-guided pruning marginally improves overall accuracy (0.47 vs. 0.43 baseline) while text-guided and combined pruning show no change, consistent with the redundancy pattern observed without fine-tuning. A longer training run with more balanced question-type coverage would be needed to draw stronger conclusions.

Notably, all pruned conditions achieve higher LLM-judge reasoning scores than the fine-tuned no-pruning baseline (2.20–2.40 vs. 2.00), and the preference evaluation shows pruned outputs are preferred over the fine-tuned baseline for text and gaze conditions (Figure 5). This suggests that pruning may interact positively with fine-tuning: the focused visual context produced by pruning appears to support more grounded reasoning in the fine-tuned model.

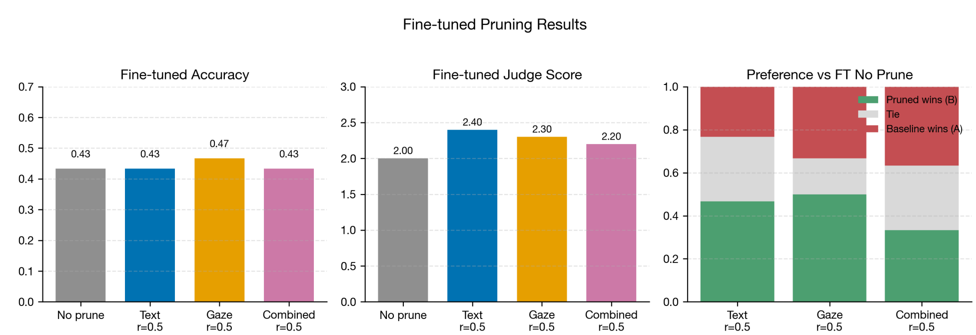

Figure 5: Fine-tuning Results showing performance in MCQ accuracy, as well as LLM judge score and LLM judge Preference between FT no prune and the 3 FT pruning categories

Qualitative before/after comparison. Table 7 shows an example of the CoT outputs the LLMJudge was evaluating on and a particular instance where the model seems to reason through the gaze shift which is an aid for the correct answer of the problem. This example among others suggest that as VLMs are fine tuned for more reasoning tasks their pruning with text and gaze will be a successful avenue to decrease their inference latency.

5 Conclusion

In this paper, we investigated weather inference-time token pruning can help to reduce decoding cost while maintaining a reasonable accuracy. Human gaze and text-prompt were used as pruning signals for VLMs on egocentric video input. For all tasks MCQ, caption similarity and CoT quality,

Table 7: Chain-of-thought comparison before and after GRPO fine-tuning, causal sample, no-pruning condition. **Q:** *Why did I move jigsaw pieces from the first box to the second box while organising them?* Correct answer: C.

Model	**Reasoning chain (truncated)**
Base model × Ans: D	*"The person is moving pieces from one box to another, which suggests they are* ***preparing them for a friend*** *who asked for them to be organised."*
GRPO fine-tuned ✓ Ans: C	*"The person is looking at the pieces, indicating purposeful action.* ***Gaze shifts suggest sorting by shape and colour****, which matches the observed behaviour."*

pruning up to 50% of visual tokens does not significantly degrade reasoning performance relative to the unpruned baseline (McNemar's test, all $p > 0.05$). This holds for both the 2B Qwen2-VL model on EgoGazeVQA and the SmolVLM prototype on EgoExo4D, confirming that substantial visual redundancy exists in current VLM decoding pipelines.

Across pruning methods, text-guided pruning at r=0.5 achieves highest captioning similarity (0.818 vs 0.815 baseline) and throughput gain (8%), while gaze-guided pruning shows accuracy frop on causal questions (0.61 vs 0.71 baseline). This is expected since gaze encodes spatial attention, not causal or relational relevance. Gaze gives a small advantage on temporal questions, but does not substitute prompt-conditioned relevance when question demands reasoning beyond the region with saliency.

The GRPO fine-tuning results provide a more nuanced view of gaze. Although the short fine-tuning run does not consistently improve overall model performance, Table 6 shows that gaze-guided pruning performs best among the fine-tuned conditions, improving overall accuracy from 0.43 to 0.47 and spatial accuracy from 0.08 to 0.15 relative to the fine-tuned no-pruning baseline. This suggests that gaze may become more useful when the model has been adapted to the egocentric reasoning task, particularly for spatially grounded questions where fixation information is more directly relevant. However, the drop in the combined model's performance suggest that combining gaze-guided and text-guided pruning is more nuanced than a linear fusion scheme. We suspect the reason for the delicate interplay between these pruning strategies is that gaze and text may encode different forms of relevance: gaze reflects the user's spatial attention, while text reflects the semantic requirements of the question. These signals can be complementary, but they can also conflict when the user is looking at one region while the question requires information from a broader scene context. Understanding how to address such conflict is an important direction for future research.

Besides that, naive signal fusion can hurt model's pruning quality. Equal pruning weight at alpha = 0.5 can dilute both gaze and prompt signals rather then composing their strengths - text relevance and spatial gaze attention are not equally informative across question types, and a fixed scalar blend cannot adapt to that variation. It is important for future work to dynamically assign alpha values based on the relative importance of gaze and prompt signals for each task.

Finally, pruning layer placement showed to matter more than pruning ratio. The layer sweep shows that pruning at layer 10 achieves 70% accuracy while layer 5 achieves only 50% — a 20-point gap using the identical keep ratio. Pruning too early discards tokens before the model has formed reliable multimodal representations, while pruning later reduces the computation savings. Layer 10 is the sweet spot for the 2B model on this task.

Overall, our findings suggest that inference-time visual token pruning is a promising direction for efficient egocentric VLM deployment. However, the results also show that gaze should not be treated as a universally reliable proxy for task relevance, and that simple gaze-text fusion is not sufficient to consistently outperform individual pruning signals. Future work should explore adaptive fusion strategies, stronger text-gaze calibration, and more extensive fine-tuning experiments on larger models and longer video contexts. More broadly, efficient egocentric VLMs will likely require pruning mechanisms that account for the user's gaze and are novelly combined with other conditional information.

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6 Course Evaluation Extra Credit

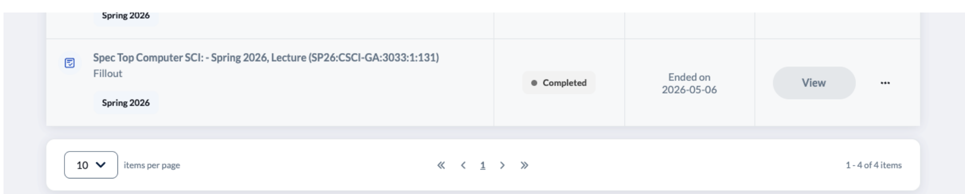

Figure 6: sara

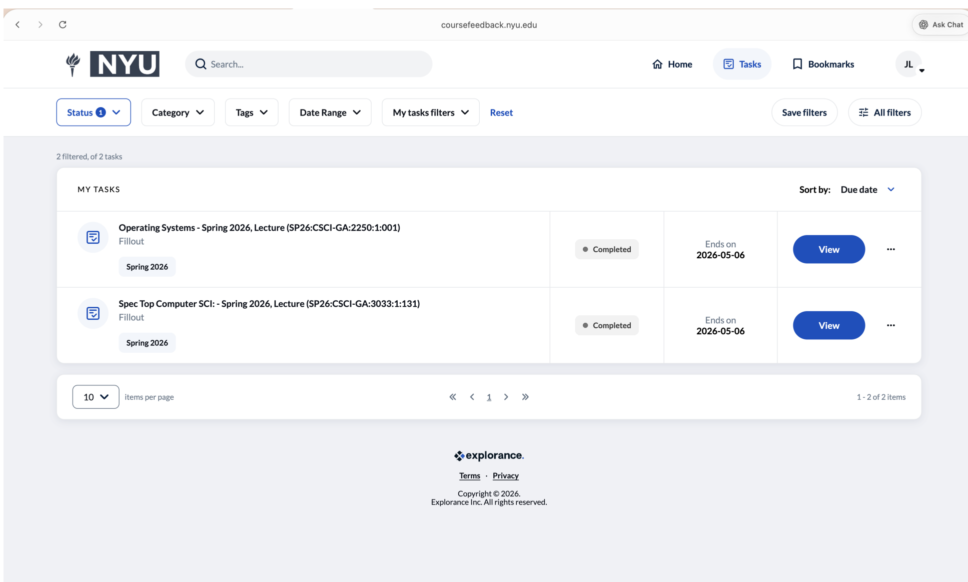

Figure 7: julia